

1. Overview & Readme

MyRecordingStudio is a dynamic web application developed using **Object-Oriented PHP and MySQL** for managing recording studio bookings across multiple locations.

The system supports two types of users:

- **Administrator**
- **Client**

Each recording studio location contains a limited number of studios and has its own hourly rental rate. Clients can view available locations, search for locations, create bookings, modify eligible bookings, cancel eligible bookings, and view their booking history.

Administrators have additional management capabilities, including managing studio locations, managing bookings on behalf of clients, viewing registered clients, and identifying clients who are currently using a studio.

The application also validates booking dates, times and durations to ensure that sessions operate within the required daily operating period of **10:00 AM to 10:00 PM**.

1.1 Requirements to Run

The application requires:

1. XAMPP with Apache and MySQL
2. PHP 7.4 or higher
3. A web browser such as Google Chrome, Microsoft Edge or Firefox
4. A code editor such as Visual Studio Code

1.2 Installation

1. Install XAMPP and open the XAMPP Control Panel.
2. Start Apache and MySQL.
3. Copy the MyRecordingStudio folder into the XAMPP htdocs folder.
4. Open phpMyAdmin: <http://localhost/phpmyadmin>
5. Select **Import**.
6. Select database/database.sql
7. Click **Go**.
8. The database myrecordingstudio will be created with the required tables and sample data.
9. Open the website: <http://localhost/MyRecordingStudio/>

2. Website Files

File	Description
index.php	Public landing page of MyRecordingStudio. Provides an introduction to the application and links to the Login and Register pages.
admin/dashboard.php	Displays the Administrator dashboard and provides access to the main administration functions.
admin/locations.php	Displays all recording studio locations and their availability. It allows the Administrator to access location management functions and view available or fully booked locations.
admin/location_form.php	Provides the form used by the Administrator to create a new location or modify an existing location.
admin/bookings.php	Displays all client bookings and allows the Administrator to manage booking sessions, including modifying eligible bookings and cancelling bookings.
admin/booking_form.php	Provides the form used by the Administrator to create a booking for a selected Client or modify an existing booking.
admin/clients.php	Displays registered Clients and Clients who are currently using a studio.
admin/search_locations.php	Allows the Administrator to search for locations using Location ID, Description or partial search terms.
auth/login.php	Provides the login form and authenticates registered Administrator and Client accounts.
auth/register.php	Allows new users to register as either an Administrator or Client.
auth/logout.php	Ends the current user session and logs the user out of the application.
client/dashboard.php	Displays the Client dashboard with an overview of the Client's booking information and links to the main Client functions.
client/book.php	Allows a Client to create a studio booking by selecting a location, booking date, session duration and available start time.
client/booking_confirmation.php	Displays the confirmation details after a successful booking, including the booking reference, session details and total cost.

client/booking_edit.php	Allows a Client to modify an eligible booking before the session has started.
client/my_bookings.php	Displays the Client's bookings, including current, upcoming, completed and cancelled sessions where applicable.
client/available_locations.php	Displays recording studio locations together with studio availability and cost information.
client/search_locations.php	Allows the Client to search for recording studio locations using Location ID, Description or partial search terms.
classes/Database.php	Manages the application's MySQL database connection using PDO. It provides a shared database connection for the application.
classes/User.php	Base User class that handles user registration, login, user information and authentication related operations.
classes/Client.php	Extends the User class and provides Client specific operations such as creating, modifying and cancelling bookings, viewing bookings and searching locations.
classes/Administrator.php	Extends the User class and provides Administrator specific operations such as managing locations, managing bookings on behalf of Clients and viewing Client information.
classes/Location.php	Handles recording studio location operations including creating, modifying, searching and listing locations and determining studio availability.
classes/Booking.php	Contains the main booking business logic including booking creation, modification, cancellation, cost calculation, availability checking, overlapping booking checks and booking status calculation.
classes/Validator.php	Provides server side validation functions for checking user input such as required fields, email, phone, date, time and numeric values.
config/config.php	Stores the database connection settings and application business rules such as operating hours and minimum and maximum booking duration. It also configures the Singapore timezone and PHP session.
includes/bootstrap.php	Loads the application configuration, helper functions and required classes before the main pages execute.
includes/functions.php	Contains shared helper functions for flash messages, formatting money, dates and times, displaying booking status badges and safely escaping output.
includes/header.php	Provides the shared page header, navigation and logged in

	user profile interface used throughout the application.
includes/footer.php	Provides the shared closing HTML structure used by the application's pages.
assets/css/style.css	Contains the application's visual styling including layout, navigation, colours, buttons, forms, tables, cards, booking interfaces, status badges and responsive page styling.
database/database.sql	Contains the SQL script for creating the myrecordingstudio database, creating the users, locations and bookings tables, and inserting sample users, locations and booking records for testing.

3. Database Design

Database name: myrecordingstudio

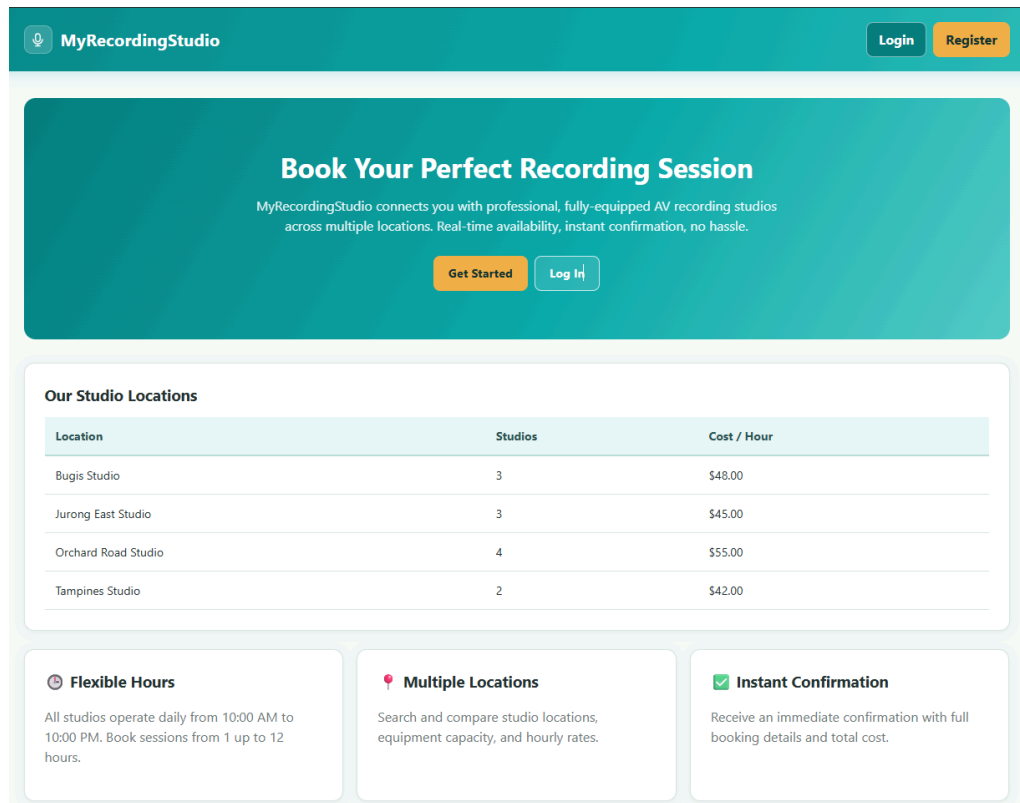
Users Table	Description
user_id	Unique user identifier
name	User's name
phone	Contact number
email	User email address
password_hash	Securely stored password
user_type	Administrator or Client
created_at	Account creation date and time

Locations Table	Description
location_id	Unique location identifier
description	Location name/description
num_studios	Number of recording studios
cost_per_hour	Studio cost per hour
created_at	Location creation date and time

Bookings Table	Description
booking_id	Unique booking identifier
client_id	Client associated with the booking
location_id	Location being booked
booking_date	Session date
start_time	Session start time
duration_hours	Booking duration
end_time	Calculated end time
total_cost	Calculated booking cost
status	Booking status such as “Upcoming”, “In Progress”, “Completed” and “Cancelled”.
created_at	Booking creation date and time

4. User Manual

4.1 Landing Page



The landing page is the first page displayed when the application is opened. The page contains the MyRecordingStudio branding, a microphone logo, a short introduction and the main Login and Register buttons.

The main section displays the message:

Book Your Perfect Recording Session

The page also displays the available studio locations with their number of studios and hourly cost. The Login button opens the login page. The Register button opens the registration page. If a user is already logged in, the application automatically redirects the user to the appropriate dashboard.

4.2 Registration

MyRecordingStudio Login Register

Create an Account

Register as a Client to book studio sessions, or as an Administrator to manage the site.

I am registering as

Full Name

Phone Number

Email Address

Password

Confirm Password

Password must be at least 6 characters.

[Already have an account? Log in here](#)

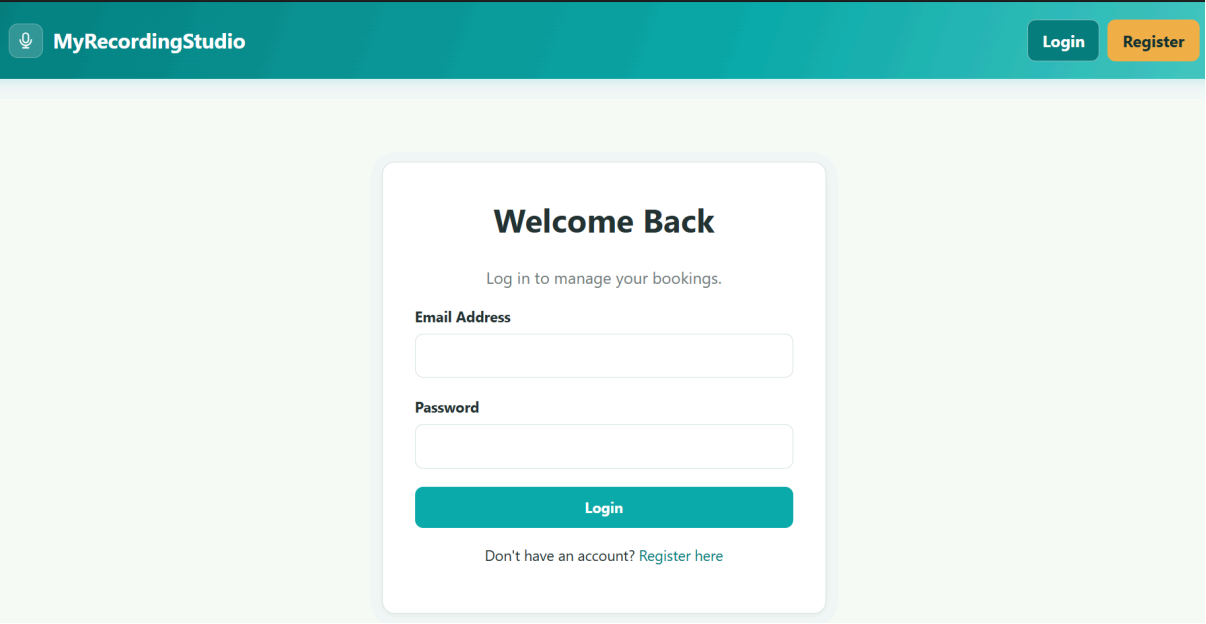
The registration page allows both Administrators and Clients to create an account.

The user enters:

1. Name
2. Phone
3. Email
4. Password
5. User type

The PHP application checks the entered information before creating the account. The email address must be valid and must not already exist in the database. The password must contain at least six characters. The password is stored using PHP password hashing rather than storing the original password.

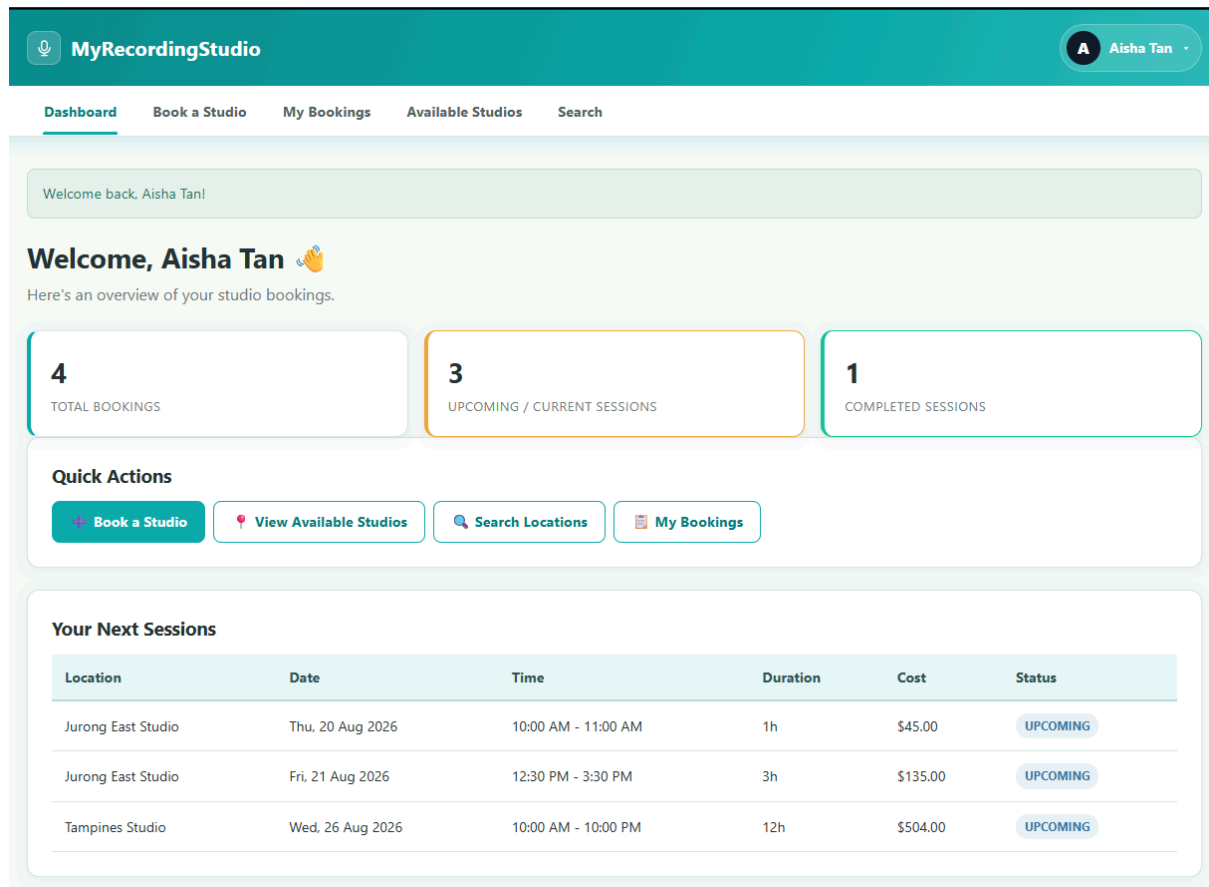
4.3 Login



The screenshot shows the login interface for 'MyRecordingStudio'. At the top, a teal header bar contains the studio's logo and name on the left, and 'Login' and 'Register' buttons on the right. The main content area has a light green background. In the center, a white card with rounded corners displays the heading 'Welcome Back' and the instruction 'Log in to manage your bookings.' Below this are two input fields: 'Email Address' and 'Password'. A teal 'Login' button is positioned below the password field. At the bottom of the card, a link reads 'Don't have an account? Register here'.

The login page allows registered users to enter their email address and password. The application checks the credentials using the User class. After successful login, the user is redirected according to the account type. An Administrator is sent to the Administrator dashboard. A Client is sent to the Client dashboard.

4.4 Client Dashboard



The Client dashboard provides a summary of the Client's booking information.

The dashboard displays:

1. Total bookings
2. Current and future sessions
3. Completed sessions

The Quick Actions section provides direct access to:

1. Book a Studio
2. View Available Studios
3. Search Locations
4. View My Bookings

This allows the Client to access the main functions without navigating through several pages.

4.5 Available Locations

MyRecordingStudio Aisha Tan

Dashboard Book a Studio My Bookings **Available Studios** Search

Locations With Available Studios

Showing locations that have at least one free studio right now.

Location	Total Studios	Studios Free Now	Cost / Hour
Bugis Studio	3	3 FREE	\$48.00
Jurong East Studio	3	3 FREE	\$45.00
Orchard Road Studio	4	4 FREE	\$55.00
Tampines Studio	2	2 FREE	\$42.00

The Available Studios page displays recording studio locations and their current availability.

The table displays:

1. Location
2. Total studios
3. Studios free now
4. Cost per hour
5. Book Now option

The application calculates the number of studios currently occupied by active bookings. For example, if a location has four studios and two studios are currently occupied, the application displays two studios as free. The **Book Now** button is an additional usability feature. When the Client selects a location, the location is automatically selected on the booking page.

4.6 Search Location

The screenshot shows the 'MyRecordingStudio' web application. The header is teal with a microphone icon and the text 'MyRecordingStudio'. On the right, there's a user profile for 'Aisha Tan'. Below the header is a navigation bar with links: 'Dashboard', 'Book a Studio', 'My Bookings', 'Available Studios', and 'Search' (which is highlighted). The main content area is titled 'Search Locations' and includes a subtitle: 'Search by Location ID and/or Description. Partial matches are supported.' Below this is a search form with two input fields: 'Location ID' (containing 'e.g. 1') and 'Description' (containing 'Orch'). A teal 'Search' button is to the right. Below the search form, there's a section titled 'Results (1)' containing a table with the following data:

ID	Description	Studios	Cost / Hour
#1	Orchard Road Studio	4	\$55.00

The application provides location search for both Clients and Administrators.

The user can search using:

1. Location ID
2. Description

The fields can also be used together. Partial search is supported.

For example, entering **Orch** can find **Orchard Road Studio**.

This directly supports the assignment requirement that users can search using only part of the search term.

4.7 Client Booking

MyRecordingStudio Aisha Tan

Dashboard **Book a Studio** My Bookings Available Studios Search

Book a Studio Session

Choose a location, date, duration and an available start time. Sessions operate from 10:00 AM to 10:00 PM.

Location
Orchard Road Studio — 4 studios — \$55.00/hr

Booking Date
15/08/2026

Session Duration (hours)
4
Choose 1 to 12 hours. The session must finish by 10:00 PM.

Available Start Times

10:00 AM 4 studios free	10:30 AM 4 studios free	11:00 AM 4 studios free	11:30 AM 4 studios free
12:00 PM 4 studios free	12:30 PM 4 studios free	1:00 PM 4 studios free	1:30 PM 4 studios free
2:00 PM 4 studios free	2:30 PM 4 studios free	3:00 PM 4 studios free	3:30 PM 4 studios free
4:00 PM 4 studios free	4:30 PM 4 studios free	5:00 PM 4 studios free	5:30 PM 4 studios free
6:00 PM 4 studios free			

Each button represents a start time that has at least one studio available for the full selected duration.

Estimated cost **\$220.00**

Confirm Booking

The Client booking page allows a Client to select:

1. Location
2. Booking date
3. Session duration
4. Available start time

The location selection displays the number of studios and hourly cost. The duration can be selected from one hour to twelve hours.

The estimated cost is updated automatically based on: $\text{Total Cost} = \text{Session Duration} \times \text{Cost Per Hour}$
For example: $4 \text{ hours} \times \$55.00 = \220.00

4.8 Available Start Times

Book a Studio Session

Choose a location, date, duration and an available start time. Sessions operate from 10:00 AM to 10:00 PM.

Location
Orchard Road Studio — 4 studios — \$55.00/hr

Booking Date
15/08/2026

Session Duration (hours)
6

Choose 1 to 12 hours. The session must finish by 10:00 PM.

Available Start Times

10:00 AM 4 studios free	10:30 AM 4 studios free	11:00 AM 4 studios free	11:30 AM 4 studios free
12:00 PM 4 studios free	12:30 PM 4 studios free	1:00 PM 4 studios free	1:30 PM 4 studios free
2:00 PM 4 studios free	2:30 PM 4 studios free	3:00 PM 4 studios free	3:30 PM 4 studios free
4:00 PM 4 studios free			

Each button represents a start time that has at least one studio available for the full selected duration.

Estimated cost

\$330.00

Confirm Booking

After the Client selects the location, date and duration, the application requests the available start times from the PHP booking logic. The available times are displayed as selectable buttons.

Each button shows:

- Start Time
- Studios Free

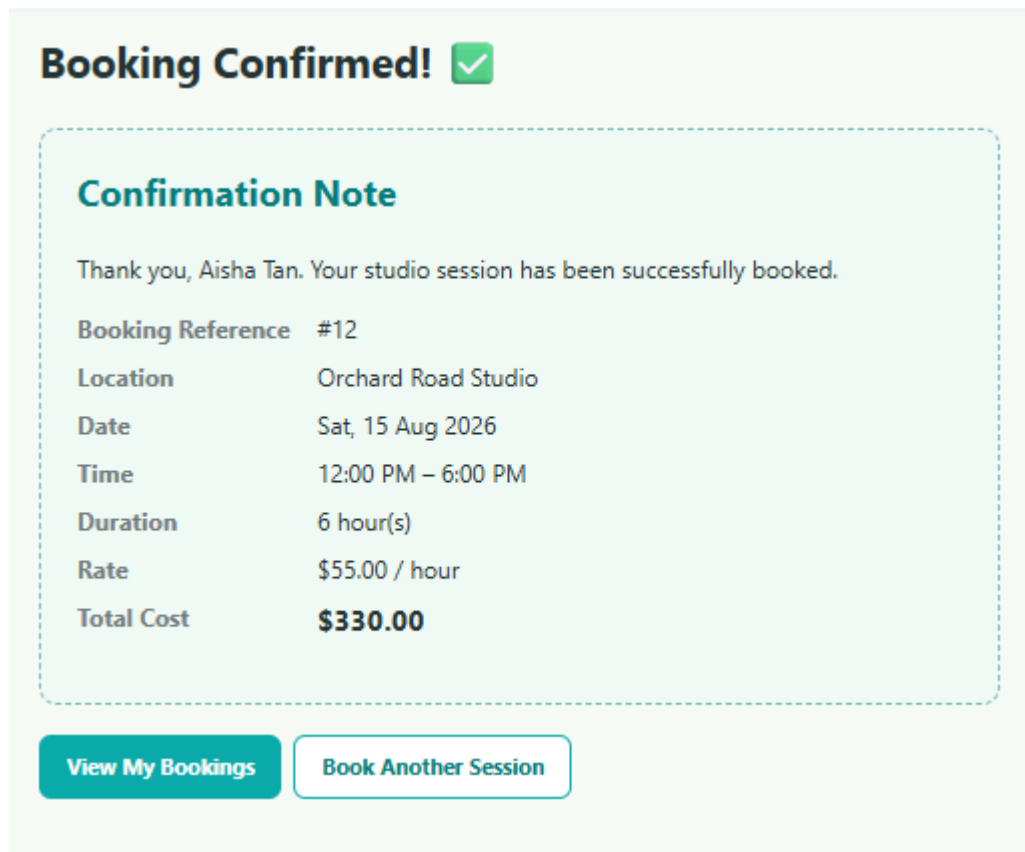
Only times where at least one studio can accommodate the complete selected duration are displayed.

The system checks the selected duration before showing the time.

For example, if the Client selects a six hour session, a start time is displayed only when the session can finish by 10:00 PM and the required studio capacity is available.

This makes the booking process easier because the Client does not need to repeatedly enter different times to find an available session.

4.9 Booking Confirmation



After a successful booking, the application displays a confirmation note.

The confirmation contains:

1. Booking reference
2. Location
3. Booking date
4. Start time
5. End time
6. Duration
7. Hourly rate
8. Total cost

The total cost is calculated automatically by the application.

4.10 My Bookings

MyRecordingStudio Aisha Tan

Dashboard Book a Studio **My Bookings** Available Studios Search

My Bookings

View your sessions and modify or cancel an upcoming booking.

All Bookings Current & Upcoming Completed

Ref	Location	Date	Time	Duration	Cost	Status	Actions
#11	Tampines Studio	Wed, 26 Aug 2026	10:00 AM - 10:00 PM	12h	\$504.00	UPCOMING	<button>Modify</button> <button>Cancel</button>
#6	Jurong East Studio	Fri, 21 Aug 2026	12:30 PM - 3:30 PM	3h	\$135.00	UPCOMING	<button>Modify</button> <button>Cancel</button>
#10	Jurong East Studio	Thu, 20 Aug 2026	10:00 AM - 11:00 AM	1h	\$45.00	UPCOMING	<button>Modify</button> <button>Cancel</button>
#12	Orchard Road Studio	Sat, 15 Aug 2026	12:00 PM - 6:00 PM	6h	\$330.00	UPCOMING	<button>Modify</button> <button>Cancel</button>
#13	Bugis Studio	Fri, 14 Aug 2026	10:30 AM - 2:30 PM	4h	\$192.00	IN PROGRESS	Locked
#1	Bugis Studio	Thu, 13 Aug 2026	4:30 PM - 8:30 PM	4h	\$192.00	COMPLETED	Locked

The Client can view bookings using three tabs:

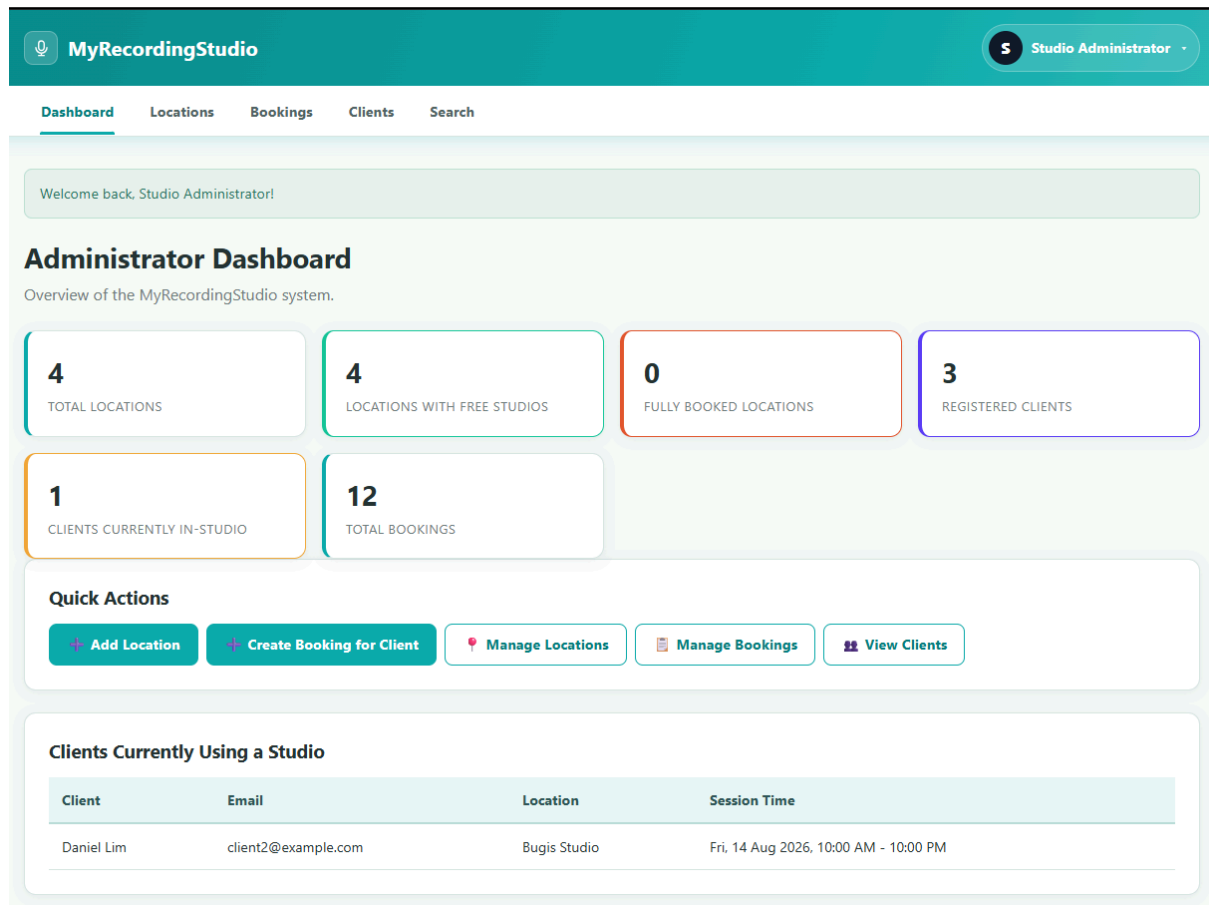
1. All Bookings
2. Current and Upcoming
3. Completed

The booking table displays:

1. Booking reference
2. Location
3. Date
4. Time
5. Duration
6. Cost
7. Status
8. Available actions

The application calculates the displayed status using the booking date and time. The Client can modify or cancel an eligible booking before the session has started.

4.11 Administrator Dashboard



The Administrator dashboard provides access to the main management functions.

The navigation contains:

- Dashboard
- Locations
- Bookings
- Clients
- Search

The Administrator also has a profile button in the top right corner. The profile menu displays the Administrator name, email address, business name, account role and Logout option.

4.12 Location Management

The screenshot displays the 'MyRecordingStudio' web application interface. The top navigation bar is teal with a microphone icon and the text 'MyRecordingStudio'. On the right, a user profile dropdown shows 'Studio Administrator'. Below the navigation bar, a menu includes 'Dashboard', 'Locations' (which is underlined), 'Bookings', 'Clients', and 'Search'. The main content area is titled 'Studio Locations' with a subtitle 'Create, edit, and monitor all recording studio locations'. A teal button with a plus icon and the text 'Add New Location' is present. Below this, there are three tabs: 'All Locations' (selected), 'With Available Studios', and 'Fully Booked'. A table lists the studio locations with columns for ID, Description, Total Studios, Cost / Hour, and Actions. Each row has an 'Edit' button in the Actions column.

ID	Description	Total Studios	Cost / Hour	Actions
#2	Bugis Studio	3	\$48.00	<button>Edit</button>
#4	Jurong East Studio	3	\$45.00	<button>Edit</button>
#1	Orchard Road Studio	4	\$55.00	<button>Edit</button>
#3	Tampines Studio	2	\$42.00	<button>Edit</button>

The Administrator can view all studio locations.

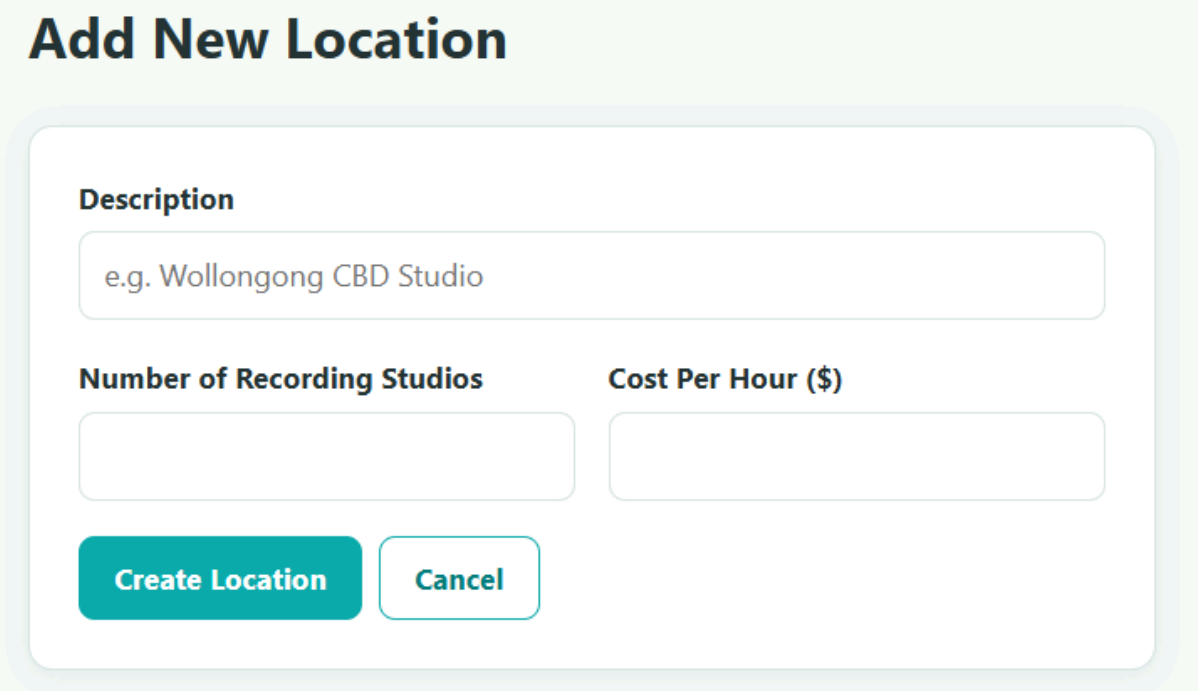
The location information includes:

1. Location
2. Total studios
3. Available studios
4. Cost per hour

The Administrator can also view locations where all studios are currently occupied.

The application determines availability using the active bookings and the number of studios stored for each location. The Administrator can modify existing location information. The updated information is stored in the locations table.

4.13 Create Location



The image shows a web form titled "Add New Location" with a light green header. The form is enclosed in a rounded rectangle with a light blue border. It contains three input fields: a text field for "Description" with a placeholder "e.g. Wollongong CBD Studio", a text field for "Number of Recording Studios", and a text field for "Cost Per Hour (\$)". Below these fields are two buttons: a teal "Create Location" button and a white "Cancel" button with a teal border.

Add New Location

Description

e.g. Wollongong CBD Studio

Number of Recording Studios

Cost Per Hour (\$)

Create Location **Cancel**

The Administrator can create a new recording studio location.

The form requires:

1. Description
2. Number of recording studios
3. Cost per hour

The PHP Location class validates the entered values before inserting the new location into the database. The number of studios must be a positive whole number. The cost per hour must be a positive number.

4.14 Administrator Booking Management

MyRecordingStudio

S Studio Administrator

Dashboard

Locations

Bookings

Clients

Search

Booking cancelled successfully.

All Bookings

Create, modify and cancel client booking sessions.

+ Create Booking for a Client

Ref	Client	Location	Date	Time	Cost	Status	Actions
#5	Daniel Lim client2@example.com	Bugis Studio	Fri, 28 Aug 2026	11:00 AM - 1:00 PM	\$96.00	CANCELLED	Locked
#11	Aisha Tan client1@example.com	Tampines Studio	Wed, 26 Aug 2026	10:00 AM - 10:00 PM	\$504.00	UPCOMING	Modify Cancel
#6	Aisha Tan client1@example.com	Jurong East Studio	Fri, 21 Aug 2026	12:30 PM - 3:30 PM	\$135.00	UPCOMING	Modify Cancel
#4	Daniel Lim client2@example.com	Tampines Studio	Fri, 21 Aug 2026	10:00 AM - 3:00 PM	\$210.00	CANCELLED	Locked
#10	Aisha Tan client1@example.com	Jurong East Studio	Thu, 20 Aug 2026	10:00 AM - 11:00 AM	\$45.00	UPCOMING	Modify Cancel
#3	Daniel Lim client2@example.com	Orchard Road Studio	Wed, 19 Aug 2026	2:30 PM - 4:30 PM	\$110.00	UPCOMING	Modify Cancel
#12	Aisha Tan client1@example.com	Orchard Road Studio	Sat, 15 Aug 2026	12:00 PM - 6:00 PM	\$330.00	UPCOMING	Modify Cancel
#13	Aisha Tan client1@example.com	Bugis Studio	Fri, 14 Aug 2026	10:30 AM - 2:30 PM	\$192.00	IN PROGRESS	Modify Cancel
#2	Daniel Lim client2@example.com	Bugis Studio	Fri, 14 Aug 2026	10:00 AM - 10:00 PM	\$576.00	IN PROGRESS	Modify Cancel
#8	Daniel Lim client2@example.com	Bugis Studio	Thu, 13 Aug 2026	5:00 PM - 10:00 PM	\$240.00	COMPLETED	Locked
#9	Mei Wong client3@example.com	Bugis Studio	Thu, 13 Aug 2026	5:00 PM - 9:00 PM	\$192.00	COMPLETED	Locked
#1	Aisha Tan client1@example.com	Bugis Studio	Thu, 13 Aug 2026	4:30 PM - 8:30 PM	\$192.00	COMPLETED	Locked

The All Bookings page allows the Administrator to manage and monitor all booking sessions made by clients. All client bookings are displayed in one table with the booking reference, client details, location, date, time, cost and current status.

The Administrator can modify eligible upcoming bookings and cancel eligible bookings when required. Cancelled and completed bookings are shown as Locked, so they cannot be modified or cancelled again.

The booking status clearly shows whether a session is Upcoming, In Progress, Completed or Cancelled. This allows the Administrator to quickly monitor the current state of every

booking. The Administrator can also select Create Booking for a Client to create a new studio booking on behalf of a specific client.

4.15 Administrator Create Booking for a Client

Create Booking for a Client

Select the client, location, date and duration. Available start times are shown automatically.

Client

-- Select a client --

Location

-- Select a location --

Booking Date

dd/mm/yyyy

Session Duration (hours)

Choose 1 to 12 hours. The session must finish by 10:00 PM.

Available Start Times

Select a client, location, date and duration to see available start times.

Estimated cost

\$0.00

Create Booking

Cancel

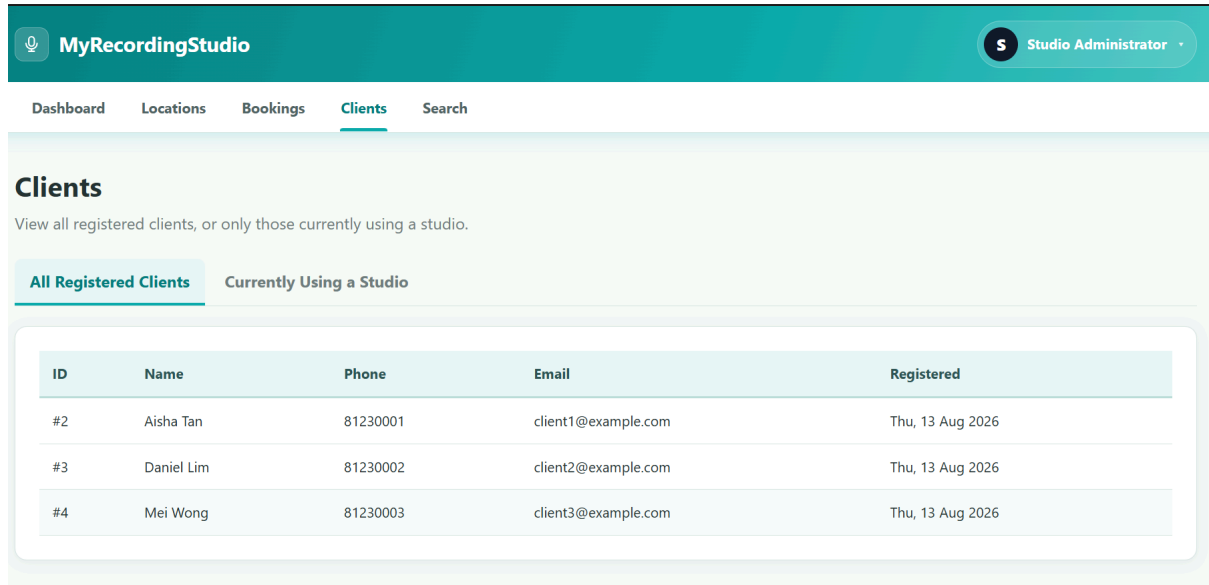
The Administrator can select:

1. Client
2. Location
3. Booking date
4. Session duration
5. Available start time

The application displays the available start times automatically. The same booking rules used by the Client booking process are applied. This prevents the Administrator from creating an invalid booking.

20

4.16 Registered Clients



ID	Name	Phone	Email	Registered
#2	Aisha Tan	81230001	client1@example.com	Thu, 13 Aug 2026
#3	Daniel Lim	81230002	client2@example.com	Thu, 13 Aug 2026
#4	Mei Wong	81230003	client3@example.com	Thu, 13 Aug 2026

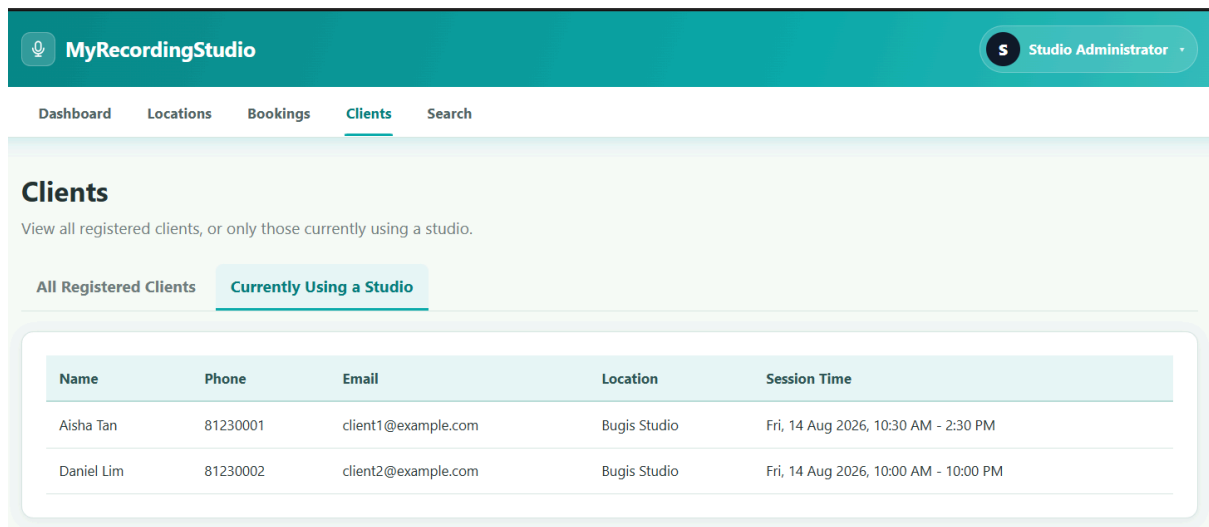
The Administrator can view all registered Clients.

The table displays:

1. Client ID
2. Name
3. Phone
4. Email
5. Registration date

This allows the Administrator to view the Clients registered in the system.

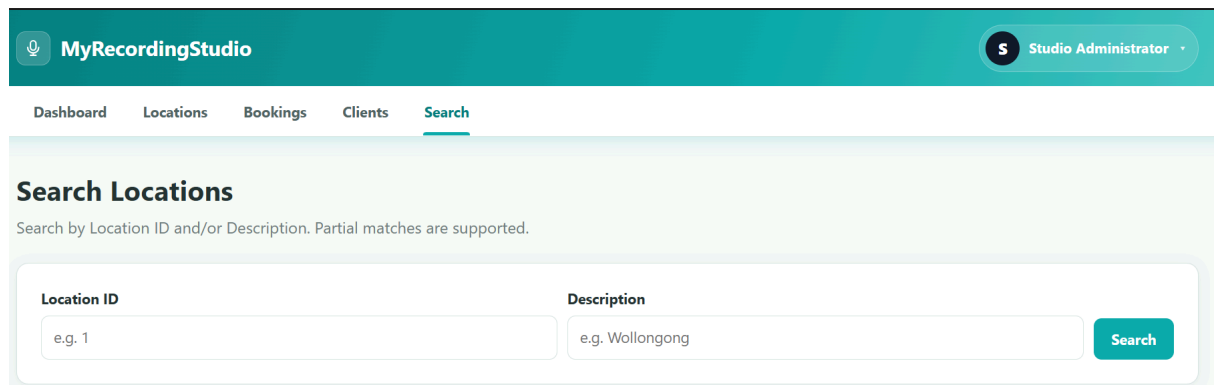
4.17 Clients Currently Using a Studio



Name	Phone	Email	Location	Session Time
Aisha Tan	81230001	client1@example.com	Bugis Studio	Fri, 14 Aug 2026, 10:30 AM - 2:30 PM
Daniel Lim	81230002	client2@example.com	Bugis Studio	Fri, 14 Aug 2026, 10:00 AM - 10:00 PM

The Administrator can view Clients who are currently using a studio. The application checks whether the current time falls between the booking start time and end time. This provides the Administrator with a current view of Clients using studio facilities.

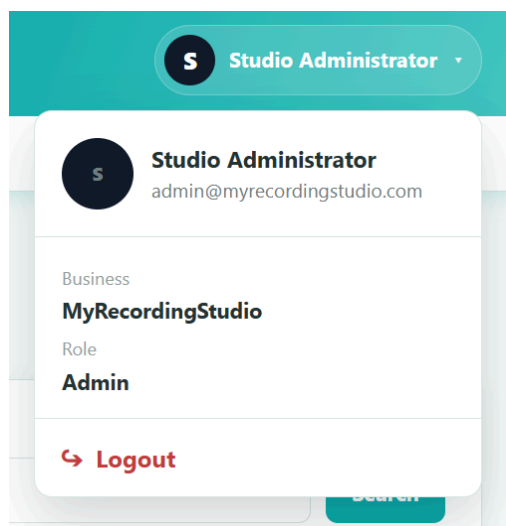
4.18 Administrator Search



The screenshot shows the 'MyRecordingStudio' web application interface for a 'Studio Administrator'. The top navigation bar includes links for 'Dashboard', 'Locations', 'Bookings', 'Clients', and 'Search'. The 'Search' tab is active. Below the navigation bar, the 'Search Locations' section is displayed. It includes a sub-header 'Search Locations' and a note: 'Search by Location ID and/or Description. Partial matches are supported.' There are two input fields: 'Location ID' with a placeholder 'e.g. 1' and 'Description' with a placeholder 'e.g. Wollongong'. A 'Search' button is located to the right of the input fields.

The Administrator can search for locations using Location ID and Description. Partial matching is supported. The search result displays the matching location information and provides access to modify the selected location.

4.19 Logout



The profile menu is available in the top right corner after login. It displays:

- User name
- Email
- Business
- Role

- Logout

Selecting Logout calls auth/logout.php. The application clears the active session and returns the user to the public landing page.